Lifting Tool for 777F Final Hub - Ring Gear Group

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

Editor's Note: The use of this installation tool and the processes discussed within this Best Practice is not a substitute for applicable laws, standards and regulations and does not alter or limit the obligation of individuals who utilize the tool to fully comply with federal, state and local law and prudent safety measures relating to the use of lifting devices.

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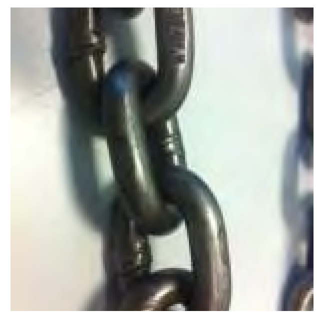
1.0 Introduction

The assembly process of the 777 final drives is a task that is performed frequently at the CRC shops. Due to this and taking into account the size and weight of the pieces to be assembled, there is an increase of the risk of serious accidents in the vertical process assembly at the shop (similar procedure to those used for the models 793, as described in the SIS web).

This Best practice aims to facilitate the development of this task thru the design a lifting tool which is fast, easy, and safe use.

2.0 Best Practice Description

This tool applies to the 777 Final drive Ring gear – Hub group. It consists of two symmetrical arms joined by a 7/16 "x 1 ½" 8 grade in the center, forming a reverse scissor, and it also has a calibrated chain link 21" long attached to each upper end of the arms by a bolt 7/16" x 1 ½" 8 grade (each bolt has its nut and washer welded).



The tool weight is 2.7 kg, and has a safety factor equal to 9.7 corresponding to the lifting weight, i.e. 200 kg 1. It´s necessary to keep in mind that for the correct lifting, the part should be held in three spots, thus, these different tools are required.
This system is used to carry the ring group from the workbench to the height of the final drive upright at approximately 1.8 m. It's worth mentioning that the final drive ring gear group can be installed unaffectedly by the use of the tool, being the tool removed once it is positioned. i.e. once the part is laid on the final drive, the internal geometry of this allows enough space for the tool not to be trapped and can be easily removed smoothly just by releasing the lifting spots.

3.0 Implementation Steps

• Check the status of the tool (1) to verify the integrity of it before be using.
• Check that the hub (2) is properly torqued to the ring gear to avoid problems during the lifting.
• Position the 3 units in each hub hole (3) so that the chains can be lifted and stressed by the jib crane hook (4).
• Raise the ring gear hub group assembly - (5) until the desired height (6).
• Place the piece in its final seat and remove the tools (7).

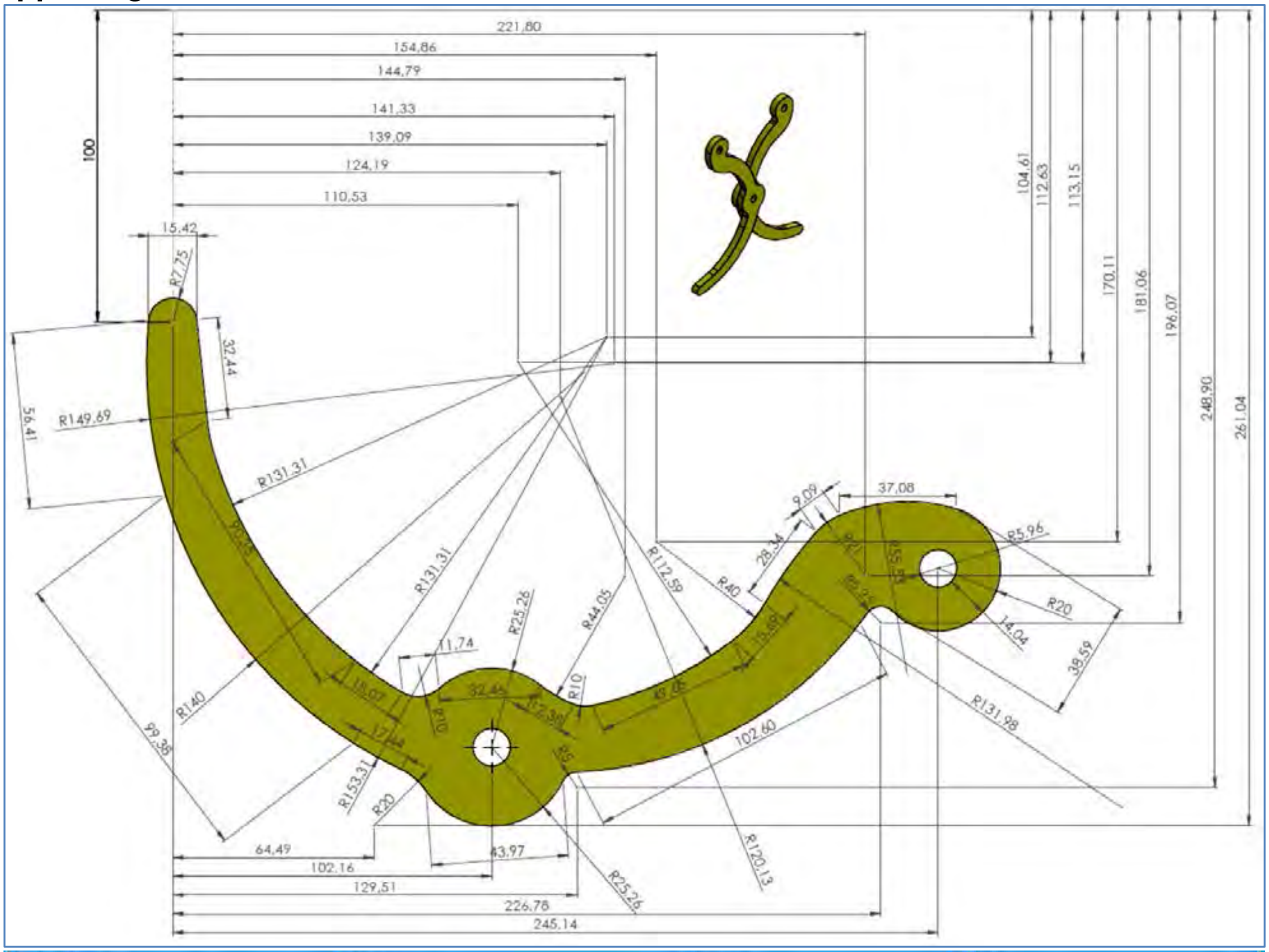
4.0 Benefits

- Easy fabrication
- Easy operation
- Improved component assembly time
- Safe load lifting

5.0 Resources Required

Drawings of the tool (see attached)

6.0 Supporting Attachments / References



Nombre	Tipo	Mín.	Máx.
Factor de seguridad1	Automático	9.70019 Nodo: 14259	1.41917e+007 Nodo: 33

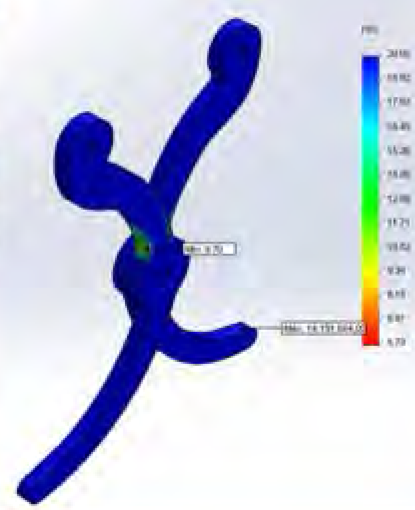
Conector de pasador/perno/rodamiento

Referencia de modelo	Detalles del conector	Detalles de resistencia
Conector tipo Pasador-1	Entidades: 2 cara(s) Tipo: Pasador Tipo de conexión: Con anillo de retención (sin traslación) Valor de rigidez rotacional: 0 Unidades: SI	Pasador sometido a esfuerzo cortante Recomendación: tornillo pasador (7/16" grado 8), arandela espaciadora y tuerca de seguridad soldada.

Fuerzas del conector

Tipo	Componente X	Componente Y	Componente Z	Resultante
Fuerza axial (N)	30.767	0	0	30.767
Fuerza cortante (N)	0	28.208	1603.5	1603.7
Torsor (N·m)	9.6957e-013	0	0	9.6957e-013
Momento flector (N·m)	0	9.6425	-0.50613	9.6558

Fuerza-1	Entidades: 1 cara(s), 1 plano(s) Referencia: PLANO2 Tipo: Aplicar fuerza Valores: ---, ---, -33Kg
Fuerza-2	Entidades: 1 cara(s), 1 plano(s) Referencia: PLANO3 Tipo: Aplicar fuerza Valores: ---, ---, -33Kg



7.0 Related Best Practices

None

8.0 Acknowledgements

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